Towards a Cubic Size Bound for POSIX Lexing with Strong Simplification:

An Incrementally Compiled Ledger of Checked Progress

Fable 5 (Claude) agent session, with the Codex (GPT-5.5) supervisor agent
repository `posix-codex`, branch `codex/backref-values`, theory `Posix` (Isabelle2025-2)

2026-06-12 – (ongoing)

Abstract

This document is an incrementally updated ledger of the campaign to prove a cubic size bound for the Antimirov-factored strong-simplification route of the POSIX lexer formalization. Every statement labelled *checked* names an Isabelle lemma that builds green in the `Posix` session; every counterexample labelled *checked* is likewise an Isabelle fact. Mirror-validated statements (random testing against a clause-verified Python model) are clearly marked as such and are *not* proofs. The document is recompiled after each increment of progress.

Contents

1 The target

For the erased regular-expression datatype `rrexp` (constructors `RZERO`, `RONE`, `RCHAR`, `RSEQ`, `RALTS`, `RSTAR`, `RNTIMES`, plus backreference constructors excluded from the fragment), the live target is the *set-ledger cubic gate* (`MAINLINE.md`):

$$\mathrm{rsize\_set}\big(\texttt{row\_dlformss}\,(\texttt{rpder\_strong\_rows\_raw}\;c\;(\texttt{afactored1}\;r\;s))\big)\;\le\;2\,(\mathrm{rsize}\,r+3)^3.$$

In words: take the rows produced by one strong derivative step from the factored front of r after input s, open every row into its linear forms, deduplicate *across the whole union*, and sum the sizes of the distinct opened rows; this total must be cubic in the size of the original regex. POSIX value preservation is handled by the separately checked deferred-memo reconstruction theorems and is not re-proved here.

2 Checked refutation: the duplicated list account is exponential

The first reduction attempt measured the *duplicated list* cost `afactored1_strong_dlform_list_cost` (sum of opened-list sizes, duplicates counted). This is provably hopeless.

Theorem 2.1 (checked: `afactored1_strong_dlform_list_cost_cubic_false`)**.** *There is a family r_j with $\mathrm{rsize}\,r_j = \Theta(j)$ whose one-step duplicated list cost grows as 2^j; in particular the cost is not bounded by $2(\mathrm{rsize}\,r + 3)^3$, nor by any polynomial.*

The witness is the *RONE-pair tower*: with $m' = \texttt{RSEQ}\,(\texttt{RALTS}[\texttt{RONE}, a])\,(\texttt{RALTS}[\texttt{RONE}, b])$ and $x_j = \texttt{RSEQ}\,(\texttt{RALTS}[\texttt{RONE}, m'])\,x_{j-1}$, every nesting level doubles the number of opened rows because the two `RONE`-paths of m' both reopen the tail; the tower is a literal fixpoint of the strong simplifier, so simplification does not help. All fully collapsing `RONE`-paths produce *the same* row, which is why the deduplicated (set-level) account below survives.

3 Checked set-level theory

Theorem 3.1 (checked: `card_row_dlforms_le_rsize`)**.** *For every row q: $|\textit{row\_dlforms}\,q| \le \mathrm{rsize}\,q$. (Distinct opened rows of a single row are at most linear, unconditionally.)*

The proof is a set-difference triangle along the reassociation chain; the core checked inequality is `card_row_dlforms_rsimp4_diff_le`: $|\texttt{row\_dlforms}(\sigma_4(p, k)) \setminus \texttt{row\_dlforms}\,k| \le \mathrm{rsize}\,p$, where σ_4 is the sequence-atom smart constructor.

Theorem 3.2 (checked: `rsize_set_row_dlforms_le_rsize_sq`)**.** $\mathrm{rsize\_set}(\textit{row\_dlforms}\,q) \le (\mathrm{rsize}\,q)^2$ *for every row q.*

Theorem 3.3 (checked: `card_row_dlformss_le_rsizes` and `card_row_dlformss_rpder_strong_rows_raw_le_generated`) *The deduplicated union over a row list has at most* $\mathrm{rsizes}(rows)$ *members; instantiated to the actual one-step output, the union card is at most the generated total size (one degree).*

4 Checked gate reduction: from three summands to two numbers

The supervisor's pair-budget gate splits the union ledger into a non-sequence part, a front-times-tail-weights part, and an active-suffix pair-budget part. Both abstract premises of the gate were discharged (`actual_union_seq_head_size_linear` gives head bound $H = \mathrm{Suc}(2\,\mathrm{rsize}\,r)$ on the legacy fragment; `rseq_tails_row_dlformss_rpder_strong_weight_le_generated` gives the key weight M), yielding the instantiated gate `actual_union_pair_budget_gate_instance`. Then:

Theorem 4.1 (checked: `pair_budget_afactored1_strong_dlform_universe_zero`)**.** *The active-suffix pair budget of the dlform universe is identically zero: opened rows always have non-alternation heads (`row_dlforms_seq_member_head_nonalt`), so no member of the universe has the keyed shape `RSEQ` (`RALTS` rows) k. The third gate summand vanishes (`actual_union_gate_two_summands`).*

Theorem 4.2 (checked: exact decomposition). *With* `rsize_set_split_rnonseq_rseq`*,* `rsize_set_rseq_members_le_he` *and* `sum_rseq_member_tails_regroup`*, the whole gate rests on exactly two numbers for the actual opened union* U*:*

$$\mathrm{rsize\_set}U \;\le\; \underbrace{\textit{nonseq part}}_{\textit{cubic, checked}} + |\texttt{rseq\_members}\,U|\cdot \mathrm{Suc}\,H \;+\; \sum_t |\texttt{rseq\_tail\_rows}\,U\,t|\cdot \mathrm{rsize}t.$$

The non-sequence part is cubic by `rsize_set_rnonseq_members_..._cubic` *via the front-terms carrier.*

5 Checked static universe results

Theorem 5.1 (checked: `rsizes_afactored1_le_rsize_set_apder_rows`). *For* every *input* s*:* $\mathrm{rsizes}(\texttt{afactored1}\,r\,s) \le \mathrm{rsize\_set}(\texttt{apder\_rows}\,r)$*. The multi-step front total is bounded by a static, input-independent quantity (rows are distinct and live in the static row universe).*

Theorem 5.2 (checked: `apder_rows_member_size_quadratic`). *On the* `apder_nf` *fragment every member of the static row universe has size at most* $\mathrm{Suc}((\mathrm{rsize}r+2)^2)$*. Hence every dynamic front row is at most quadratic — the row-size half of the front-quadratic conjecture.*

Counterexample 5.3 (mirror-verified, tight). *The static ledger* $\mathrm{rsize\_set}(\texttt{apder\_rows}\,r)$ *is* cubic-tight*: on the star-seq towers* $x_{i+1} = (x_i\,b_i)^*$ *the ratio* $\mathrm{rsize\_set}/(\mathrm{rsize})^3$ *converges to* ≈ 0.040*. The checked cubic bound* `rsize_set_apder_rows_expanded_cubic_size_bound` *cannot be improved to quadratic. Moreover the* reachable *row universe equals the static one on this family, so no set-shaped (history-independent) bound can beat cubic; the missing degree is irreducibly about rows* coexisting *in one front (liveness).*

6 The row-count law (the "D law")

The remaining static improvement is row-*count* linearity, classical Antimirov shape. Write $\mathrm{acc}rk$ for `apder_term_frontier_acc`, $\mathrm{F}k$ for `rfrontier` k.

Definition 6.1 (checked: `apder_zw2`). $\mathrm{zw}_2(\texttt{RCHAR}) = 1$*; sums over* `RALTS`*/*`RSEQ`*;* $\mathrm{zw}_2(\texttt{RSTAR}\,x) = 1+\mathrm{zw}_2(x)$*;* $\mathrm{zw}_2(\texttt{RNTIMES}\,r\,n) = n\,(1+\mathrm{zw}_2 r)$*; 0 otherwise. Checked:* $\mathrm{zw}_2 \le \mathrm{rsize}$ *on the* `rntimes_free` *fragment (*`apder_zw2_rntimes_free_le_rsize`*).*

Conjecture 6.2 (the D law, `rntimes_free` form; mirror: $>$ 295,000 deep samples, zero violations). *On the legacy,* `apder_nf`*,* `rntimes_free` *fragment:*

$$|\mathrm{acc}rk \setminus \mathrm{F}k| \;\le\; \mathrm{zw}_2\,r.$$

6.1 Checked falsifications along the way

Counterexample 6.3 (checked and mirror; weight too small). *The plain letter-count (*`apder_awidth`*) version is false: zero-width stars produce frontier rows without letters. The first corrected weight (*`apder_zwidth`*, star* $= \max 1$*) is* also *false at depth:* $r = b\,(\texttt{ALTS}[(a^*)^*, (b^*)^*, b])$*,* $k = \texttt{RONE}$ *gives* $5 > 4$ *— each nested zero-consuming star layer needs its own slot.*

Counterexample 6.4 (checked: `apder_zw2_D_law_rntimes_zero_alt_false`). *The unrestricted* zw_2 *law is false with zero counted repetitions:* $r = c\,(\texttt{ALTS}[\texttt{RNTIMES}\,a\,0, \texttt{RNTIMES}\,b\,0])$ *has two imported frontier atoms but* $\mathrm{zw}_2 r = 1$*. Hence the* `rntimes_free` *restriction in Conjecture 6.2.*

Nine inductive strengthenings (subset/intersect discounts, membership companions, potential and reserve forms, an undeduplicated list-length form, and the first joint invariant J^*) were falsified by explicit counterexamples; the full list with witnesses is maintained in `MATHPROBLEM_ROWCOUNT.md`. A hard methodological lesson is recorded there: two false statements each passed $\geq 95{,}000$ *shallow* random samples; all sampling now requires depth ≥ 5 plus directed nested-star/zero-width families.

6.2 Constructor bridges (checked, supervisor)

The current attack decomposes the D law by head constructor. Checked bridges so far include the `RSEQ` frontier singleton shifts, the σ_4 non-`RONE` preservation and frontier-shape lemmas, the normal-nonalt frontier/carry-bucket shifts, the right-nonalt SEQ D bridge, and its `RCHAR`-left specialization (`card_apder_term_frontier_acc_RSEQ_RCHAR_diff_le_if_right_nonalt_diff`): an ordinary D bound for an `apder_nf` non-alternation right child yields the exact parent SEQ bound when the left child is a character.

6.3 Payoff when landed

$|\texttt{apder\_rows}\, r| \leq \mathrm{zw}_2 r{+}2$, hence with the checked quadratic member bound: $\mathrm{rsizes}(\texttt{afactored1}\, r\, s) \leq (\mathrm{zw}_2 r + 2) \cdot \mathrm{Suc}((\mathrm{rsize} r + 2)^2)$ — a *cubic static front bound by assembly* on the fragment, replacing the tight-ledger detour, and the row-count half of the dynamic front-quadratic conjecture.

7 The liveness frontier

All single-step syntactic quantities are now known cubic-tight (static ledger, per-member size, per-row norm ledgers — each with explicit witness families). The remaining degree for the full gate is the *liveness slice*: simultaneously live rows are a quadratic slice of the cubic static universe. Mirror-validated with zero violations across all families is the cap-law invariant

$$\mathrm{rsizes}(\texttt{afactored\_step}\, c\ \mathit{rows}) \ \leq\ \max\bigl(\mathrm{rsizes}\, \mathit{rows},\ C\, (\mathrm{rsize} r)^2\bigr),$$

factored into three single-step obligations (jump bound, monotone self-preservation, jump saturation) — the first formulation in which every obligation quantifies over one front and one step. Formalization has not started.

8 Progress log

When	Increment
2026-06-12	Checked: atom multiplicity, double counting, per-front-row payment interface. Checked refutation: RONE-pair tower kills the duplicated list-cost target (exponential).
2026-06-12	Checked: set-level per-row theory (card linear, ledger quadratic); union card account; weight and head bounds; pair-budget gate instantiated, then its budget proven zero; gate cut to two summands; exact two-number decomposition.
2026-06-12	Checked: front staticized ($F \leq A$ for all inputs); static maxrow quadratic. Mirror: static ledger cubic-tight; reachable = static; cap-law invariant zero violations; liveness reduced to three single-step obligations.

When	Increment
2026-06-12/13	Row-count law campaign: weights `awidth`/`zwidth` falsified (checked CEs), `zw2` landed; nine strengthenings falsified; deep-sampling standard instituted; `rntimes_free` target fixed. Supervisor: constructor-bridge decomposition under way (RCHAR-left SEQ bridge checked).
2026-06-13	This document created; compiled with pdflatex.